Text Analysis

MSIS 521
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Agenda

- Similarity between documents
- Clustering
  - K-means Clustering
  - Hierarchical Clustering
- LDA Topic Modeling
- Sentiment Analysis

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Similarity and Clustering

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How Similar Are These Documents?

- He is a nice guy
- The instructor is a nice guy
- The TA is really good
- I love my TA

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How Similar Are These Documents?

- Mathematical Similarity

docs	t1	t2	t3
D1	1	0	1
D2	1	0	0
D3	0	1	1
D4	1	0	0
D5	1	1	1
D6	1	1	0
D7	0	1	0
D8	0	1	0
D9	0	0	1
D10	0	1	1

docs	t1	t2	t3
D1	2	0	3
D2	1	0	0
D3	0	4	7
D4	3	0	0
D5	1	6	3
D6	3	5	0
D7	0	8	0
D8	0	10	0
D9	0	0	1
D10	0	3	5

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Setup

- D is a collection, which contains m documents

$$D=\{d_1, d_2,\ldots d_i\ldots d_m\}\ i=1,2\ldots m$$

- There are n words among m documents.

$$di=\{w_{i1},w_{i2},\ldots w_{ij},\ldots w_{in}\}\ i=1,2\ldots m\ ,\ \ j=1,2,\ldots.n$$

- Q is a benchmark document

$$Q= \{q_1,q_2,\ldots q_i,\ldots.q_n\}\ i=1,2,\ldots.n$$

- If similarity(q, d_i) > similarity(q, d_j), then document d_i is more relevant than d_j

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Similarity Measures

- A **similarity measure** can represent the similarity between two **vectors**.
- A similarity measure is a **function** which computes the **degree of similarity** between a pair of vectors.
- Note: There are a large number of similarity measures proposed in the literature, and the measure we use depends on the purpose of analysis.

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Classic Similarity Measures

- Similarity measures are mapped to the range [-1, 1] or [0, 1]
- 0 or -1 shows minimum similarity
- 1 shows maximum similarity
- Simple normalization through Min-Max Scaling can be used to achieve the range:
  - e.g.: $s' = (s - \min_s)/(\max_s - \min_s)$

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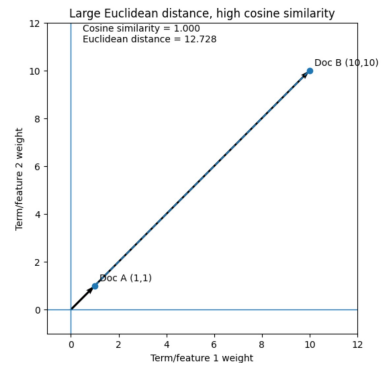
Math Expressions for Similarity Measures

- Assuming that Q and D are the sets of terms associated with a Query and a Document:
  - Simple matching $|Q \cap D|$
  - Dice's Coefficient $2\frac{|Q \cap D|}{|Q|+|D|}$
  - Jaccard Coefficient $\frac{|Q \cap D|}{|Q \cup D|}$
  - Cosine Coefficient $\frac{|Q \cap D|}{|Q|^{1/2} \times |D|^{1/2}}$
  - Overlap Coefficient $\frac{|Q \cap D|}{\min(|Q|,|D|)}$



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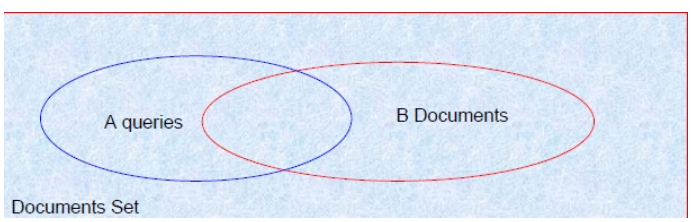
Overlap Coefficient

- Easier for vector manipulations



$$sim(q,d_j) = O(A,B) = \frac{|A \cap B|}{\min(|A|,|B|)}$$

$$\cong \frac{\sum_{k=1}^{n} w_{kq} w_{kj}}{\min(\sum_{k=1}^{n} w_{kq}^2, \sum_{k=1}^{n} w_{kj}^2)}$$

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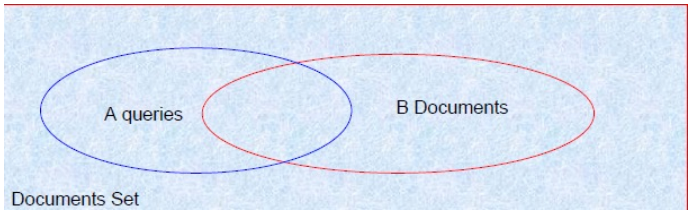
Jaccard Coefficient



$$sim(q,d_j) = J(A,B) = \frac{|A \cap B|}{|A \cup B|}$$

$$\cong \frac{\sum_{k=1}^{n} w_{kq} w_{kj}}{\sum_{k=1}^{n} w_{kq}^2 + \sum_{k=1}^{n} w_{kj}^2 - \sum_{k=1}^{n} w_{kq} w_{kj}}$$

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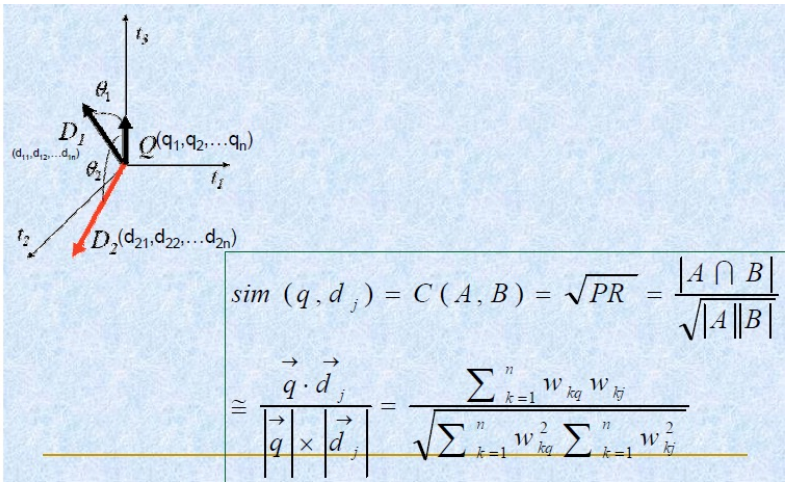
Cosine Coefficient



$$sim(q,d_j) = C(A,B) = \sqrt{PR} = \frac{|A \cap B|}{\sqrt{|A||B|}}$$

$$\cong \frac{\vec{q} \cdot \vec{d}_j}{|\vec{q}| \times |\vec{d}_j|} = \frac{\sum_{k=1}^{n} w_{kq} w_{kj}}{\sqrt{\sum_{k=1}^{n} w_{kq}^2 \sum_{k=1}^{n} w_{kj}^2}}$$

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When to Use Each Measure

- **Dice's Coefficient**: Use when you want to measure overlap but want to weight shared items a bit more than Jaccard, often in short texts or n-gram matching (e.g., in near-duplicate detection). It tends to give higher scores than Jaccard for the same overlap.
- **Jaccard Coefficient**: Use when you want a clean "intersection over union" notion of similarity on sets (unique tokens), especially when you care about distinct terms rather than how often they occur. It penalizes extra, non-overlapping terms more than Dice.
- **Cosine Coefficient**: Use when you represent documents as vectors with weights and want similarity that's robust to document length. This is the default for bag-of-words counts, TF-IDF, topic vectors, embeddings. It compares orientation, not magnitude, meaning two docs can be similar even if one is much longer, as long as term distribution is similar.
- **Overlap Coefficient**: Use when you care about containment: is the smaller document (or its phrases) largely contained in the larger? It equals 1.0 if the smaller set is a subset of the larger, even if the larger has lots of extra content. One use case is plagiarism or quote detection where one doc may include the other plus extra.

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Practice

```
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.metrics import *
from sklearn.metrics.pairwise import *

vectorizer5 = CountVectorizer()
vectorizer5.fit(c3)
print(vectorizer5.vocabulary_)
v_c3 = vectorizer5.transform(c3).toarray()
print(v_c3)

cosine_similarity([v_c3[0]],[v_c3[1]])
jaccard_score(v_c3[0],v_c3[1])
```

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Problem with Similarity

- Handling one document each time
- What if we want to handle all of the documents at once?

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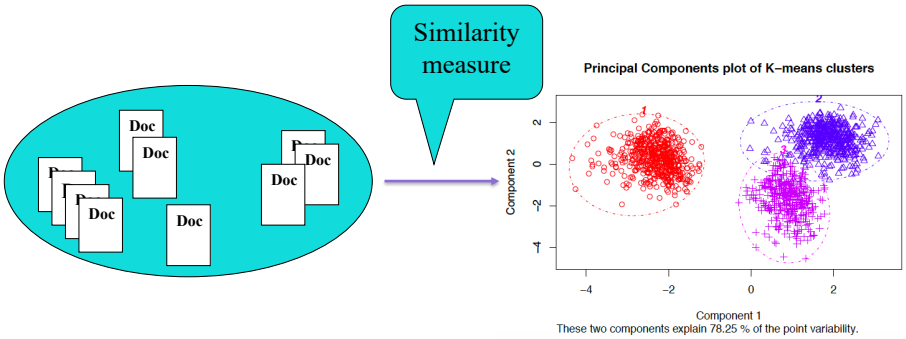
Clustering

- **Given**: Similarity measures between vectors (documents)
- **Find**: Several clusters of vectors (documents) that are similar to each other



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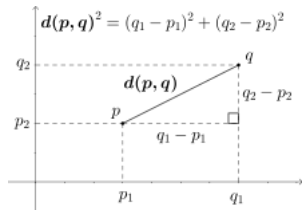
Euclidean Distance



$$dis(q, d_j) = d_E(q, d_j) = \sqrt{\sum_{k=1}^{n} (w_{kq} - w_{kj})^2}$$

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Clustering Techniques

- Partitioning Methods
- Hierarchical Methods

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Partitioning Algorithms

- **Partitioning method:** Construct a partition of ***n*** documents into a set of ***k*** clusters
- **Given:** a set of documents and the number ***k***
- **Find:** a partition of ***k clusters*** that optimizes the chosen partitioning criterion
  - **Global optimal**: exhaustively enumerate all partitions
  - Heuristic methods: **K-Means** algorithm
  - **K-Means**: Each cluster is represented by the center of the cluster

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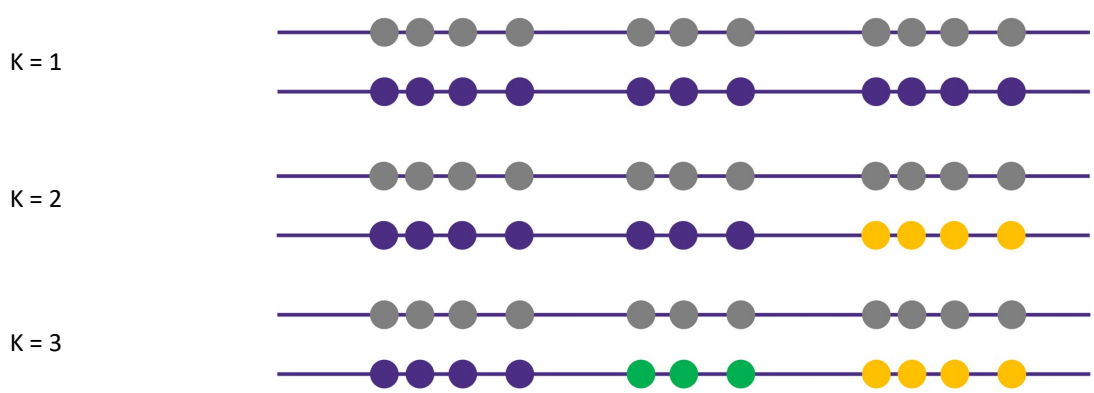
K-Means Clustering



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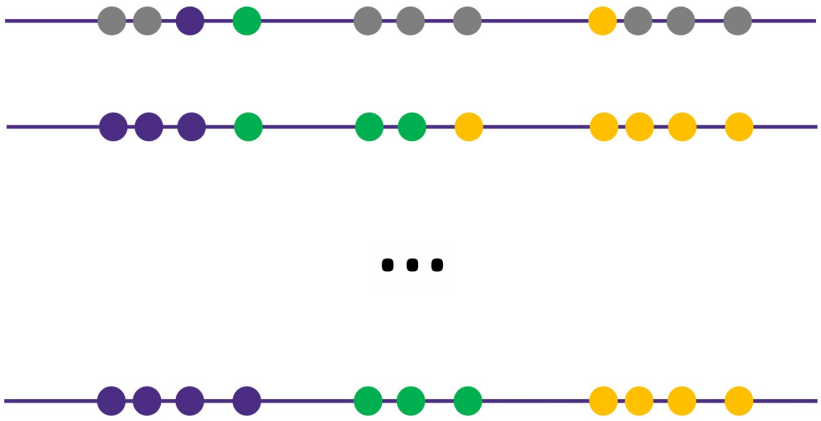
K-Means Clustering

K = 3

- Select random 3 dots
- For each non-selected dot, measure distance to the selected 3 dots, and assign it to the closest one.
- Calculate the mean (centroid) of each cluster;
- Re-measure the distance between each dot and the 3 new centroids, and reassign the clusters;
- Compute the total variation
- Try a new set of random 3 initial dots



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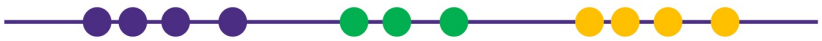
How Does the Algorithm Know the Best Clustering for a Given K?

1. Full convergence
2. Minimize the sum of the squared distance between centroid and each member of the cluster (Within Cluster Sum of Squares)
3. Max iterations reached



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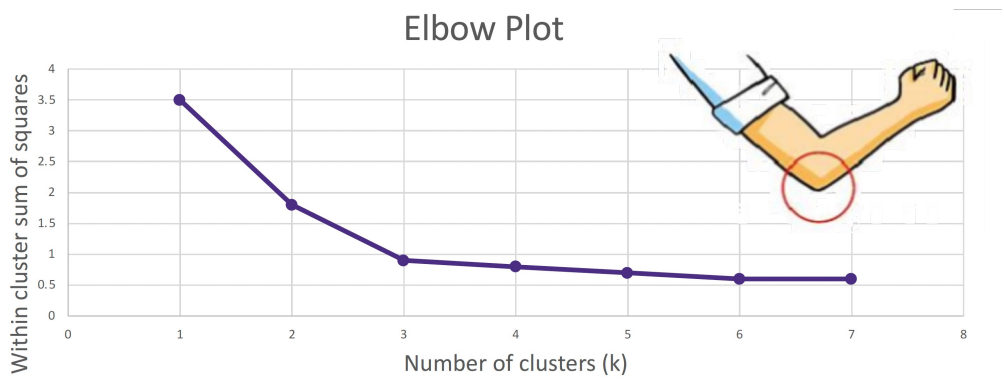
How to Select K Using the Elbow Method?



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Practice

```
c4 = [d0,d1,d2,d3,d4,d5,d6,d7,d8,d9]

vectorizer6 = TfidfVectorizer(stop_words='english')
X = vectorizer6.fit_transform(c4)

true_k = 4
model1 = KMeans(n_clusters=true_k, random_state=12, n_init=10)
model1.fit(X)
model1.cluster_centers_
model1.transform(X)
model1clusters = model1.labels_.tolist()
print(model1clusters)

order_centroids = model1.cluster_centers_.argsort()[:, ::-1]
terms = vectorizer6.get_feature_names_out()
for c in range(0, 4):
    print(f'Cluster {c}:')
    for ind in order_centroids[c, :4]:
        print(terms[ind])
```

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Hierarchical Clustering

- Agglomerative:
  - Start with each document being a single cluster.
  - In each step, merge the two closest clusters.
  - Eventually all document belong to the same cluster.
  - Objective function: minimizes the sum of squared distances within all clusters.
- Divisive:
  - Start with all documents belonging to the same cluster.
  - Eventually each node forms a cluster on its own.
  - In each step, separate the two farthest potential clusters

- Note: does not require the number of clusters k in advance

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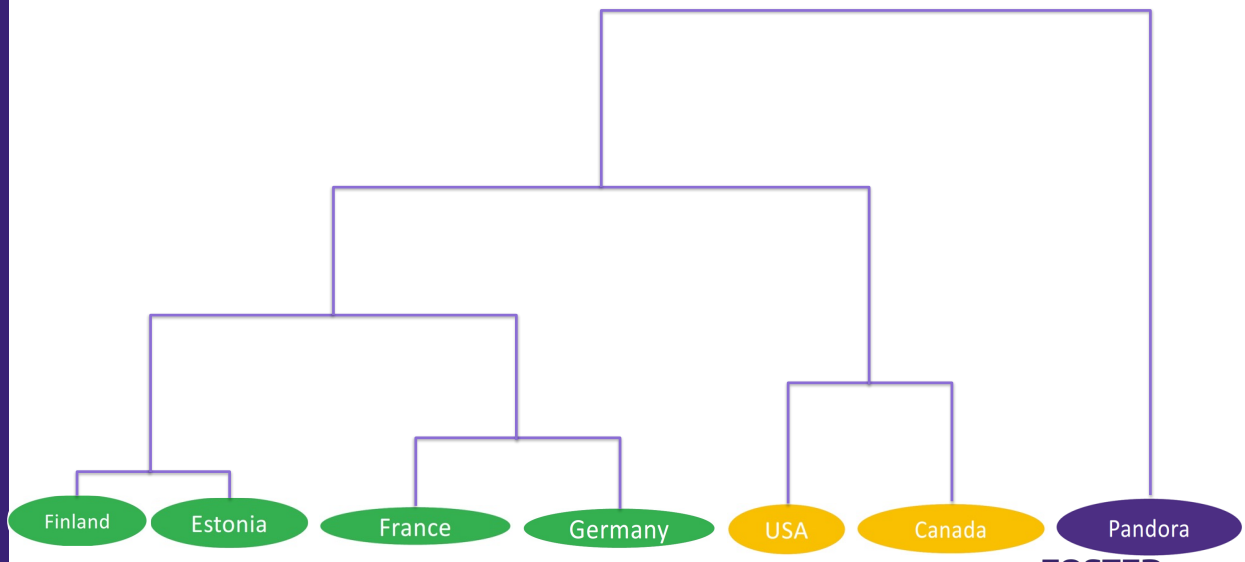
Flat vs. Hierarchical Clustering



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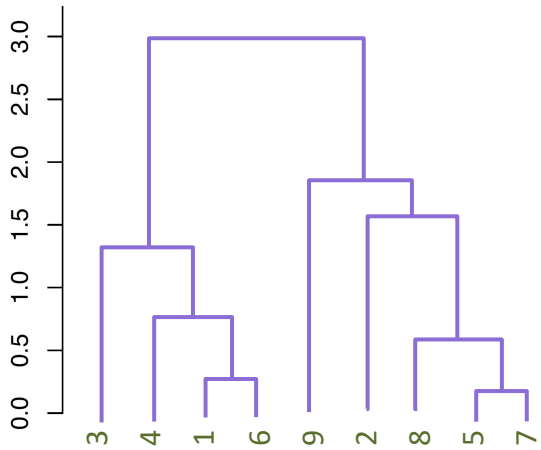
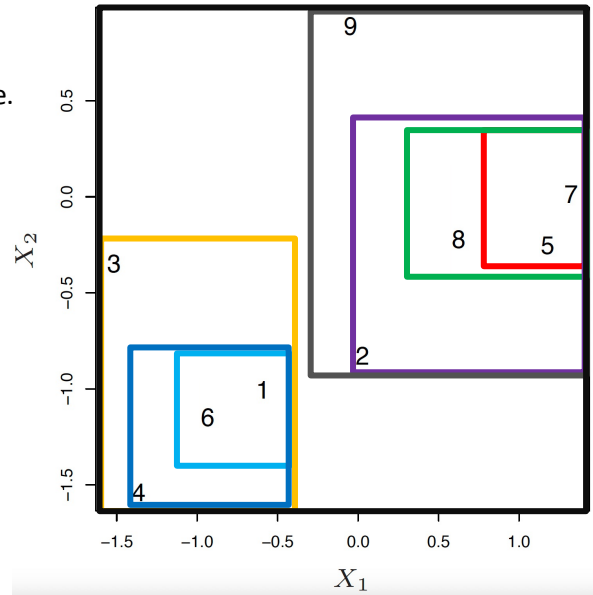
How Does the Dendrogram Work?

- A dendrogram basically keeps track of all the fuses.
- Height of fusing indicates how similar the points are.





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Practice

```
## Hierarchical clustering

from sklearn.cluster import AgglomerativeClustering

model2 = AgglomerativeClustering(n_clusters=4,
                  metric='euclidean', linkage=complete’)

model2.fit_predict(X.toarray())
print(model2.labels_)
```

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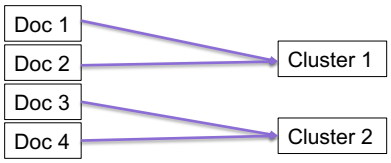
Limitations of Clustering

- Fundamental assumption: each doc belongs to one cluster, many to one relationship



- What if a doc might belong to multiple clusters?
  - E.g., can a movie (review) be related to both action and horror?
- Question: When a doc appears to belong to multiple clusters, how do we model this?

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Topic Modeling:
Latent Dirichlet Allocation (LDA)

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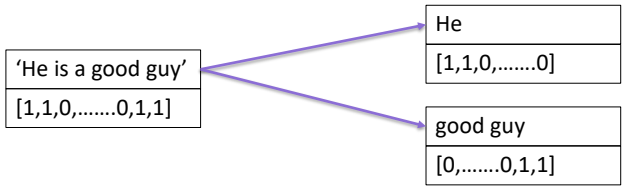
Topic Modeling

- **One data point** (vector) is part of **multiple (many) topics**
- **One topic** corresponds to specific **values** in the feature vector

- **One document** is part of **multiple (many) topics**
- **One** topic consists of **multiple** regularly co-occurring words (example topics: "sports", "medicine", "movies", "finance")



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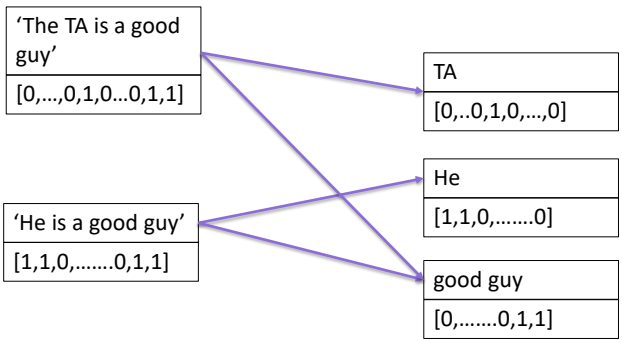
Topic Modeling

- A doc can be better described by many topics
  - Many to many relationship
- How to do that?
  - Multiple methods, but LDA is the most popular one



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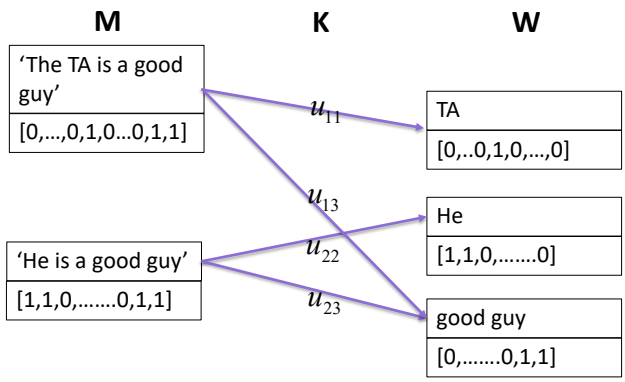
Latent Dirichlet Allocation (LDA)

- Ref: Latent Dirichlet Allocation by Blei, Ng and Jordan
  - Documents are produced from a mixture of topics
  - Topics generate words based on a probability distribution (generative probability model)



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Matrix Factorization

- General idea: perform a **low-rank approximation** of **document-term matrix**
- **M** = **KW**
  - **M**: document-term matrix
  - **K**: P(topic *t* | document *d*): each document's topic weight distribution
  - **W**: P(word *w*| topic *t*): each topic's words distribution

	W1	W2	W3	Wn
D1	0	2	1	3
D2	1	4	0	0
D3	0	2	3	1
Dn	1	1	3	0

=

	K1	K2	K3	K
D1	1	0	0	1
D2	1	1	0	0
D3	1	0	0	1
Dn	1	0	1	0

✖

	W1	W2	W3	Wm
K1	0	1	1	1
K2	1	1	1	0
K3	1	0	0	1
K	1	1	0	0

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Practice: Use sklearn

```
from sklearn.feature_extraction.text import CountVectorizer
vectorizer7 = CountVectorizer(stop_words='english')
X2 = vectorizer7.fit_transform(c4)
terms = vectorizer7.get_feature_names_out()

from sklearn.decomposition import LatentDirichletAllocation
lda = LatentDirichletAllocation(n_components=4, random_state=12).fit(X2)

for topic_idx, topic in enumerate(lda.components_):
    print(f'Topic {topic_idx}:')
    print(" ".join([terms[i] for i in topic.argsort()[:-4-1:-1]]))

lda.transform(X2[0])
```

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Practice: Use gensim

```
import gensim
dictionary = gensim.corpora.Dictionary(processed_c4)
bow_c4 = [dictionary.doc2bow(doc) for doc in processed_c4]
print(bow_c4[0])

lda_model = gensim.models.LdaModel(bow_c4, num_topics=4, id2word=dictionary)

for idx, topic in lda_model.print_topics(-1, 4):
    print('Topic: {} \n Words: {}'.format(idx, topic))

lda_model[bow_c4[0]]
```

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Sentiment Analysis

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Classification in a Descriptive Way: Sentiment Analysis

- If we have a classifier already
  - E.g., a dictionary that includes a sentiment score for each word
- Use the classifier to calculate a sentiment analysis score for documents
- He is good -> [0,0,1,0,0,0] -> score = 1/3 -> **positive**
- He is bad and crazy -> [0,0,0,-1,0,-0.5] ->score = -1.5/5 -> **negative**

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Sentiment Analysis Example 1: Positivity and Negativity

```
from nltk.sentiment.vader import SentimentIntensityAnalyzer
import pandas as pd
nltk.download('vader_lexicon')

sentences = [”They are smart, cute, and funny.",  # positive sentence example
    “They are smart, cute, and funny!", # punctuation emphasis handled correctly (sentiment intensity
adjusted)
    “They are very smart, cute, and funny.",# booster words handled correctly (sentiment intensity adjusted)
    "They are VERY SMART, cute, and FUNNY.",  # emphasis for ALLCAPS handled
    “They are VERY SMART, cute, and FUNNY!!!",  # combination of signals - VADER appropriately adjusts
intensity
    “They are VERY SMART, really handsome, and INCREDIBLY FUNNY!!!",  # booster words & punctuation make this
close to ceiling for score
    "The book was good.",  # positive sentence
    "The book was kind of good.",  # qualified positive sentence is handled correctly (intensity adjusted)
    "The plot was good, but the characters are uncompelling and the dialog is not great.",  # mixed negation
sentence
    "A really bad, horrible book.",  # negative sentence with booster words
    "At least it isn't a horrible book.",  # negated negative sentence with contraction
    ":) and :D",  # emoticons handled
    "",  # an empty string is correctly handled
    "Today sux",  # negative slang handled
    "Today sux!",  # negative slang with punctuation emphasis handled
    "Today SUX!",  # negative slang with capitalization emphasis
    "Today kinda sux! But I'll get by, lol"  # mixed sentiment example with slang and contrastive conjunction on
"but"
    ]
```

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Sentiment Analysis Example 2: Positivity and Negativity

```
## Apply existing classifier

sid = SentimentIntensityAnalyzer()

scores = []
for sentence in sentences:
    score = {'sentence': sentence}
    score.update(sid.polarity_scores(sentence))
    scores.append(score)

df_scores = pd.DataFrame(scores)
df_scores
```

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More Advanced Models

- RNN: Recurrent Neural Network
- LSTM: Long Short-Term Memory

- Transformer models
  - BERT: Bidirectional Encoder Representations from Transformers
  - ChatGPT: Generative Pretrained Transformer

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